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Reproductive performance of lactating dairy cows with elevated milk β-hydroxybutyrate levels during first 6 weeks of lactation

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ABSTRACT

Although there is evidence that ketosis negatively affects fertility, the effect of late and early ketosis on the reproductive performance of lactating cows has not been systematically investigated. The aim of this study was to evaluate the association between time and amplitude of elevated milk BHB (EMB) occurring within 42 d in milk (DIM) and subsequent reproductive performance of lactating Holstein cows. The dairy herd information data of 30,413 cows with 2 test-day milk BHB recordings during early lactation periods 1 and 2 (5–14 and 15–42 DIM, respectively) assessed as negative (<0.15 mmol/L), suspect (0.15–0.19 mmol/L), or positive (≥0.2 mmol/L) for EMB were used in this study. Based on the time and amplitude of milk BHB, cows were grouped into 7 groups: (1) healthy cows negative in both periods 1 and 2 were classified as NEG; (2) suspect in period 1 and negative in period 2: EARLY_SUSP; (3) suspect in period 1 and suspect/positive in period 2: EARLY_SUSP_Pro; (4) positive in period 1 and negative in period 2: EARLY_POS; (5) positive in period 1 and suspect/positive in period 2: EARLY_POS_Pro; (6) negative in period 1 and suspect in period 2: LATE_SUSP; and (7) negative in period 1 and positive in period 2: LATE_POS. The overall prevalence of EMB within 42 DIM was 27.4%, with the highest prevalence being EARLY_SUSP (10.49%). Cows in EARLY_POS and EARLY_POS_Pro, but not other EMB categories, had a longer interval from calving to first service compared with NEG cows. For the reproductive parameters, first service to conception interval, days open and calving interval, cows in all EMB groups except EARLY_SUSP had longer intervals compared with NEG cows. These data indicate that there is a negative association between EMB within 42 d and reproductive performance after the voluntary waiting period. The intriguing findings of this study are the unaltered reproductive performance of EARLY_SUSP cows, and the negative association between late EMB and reproductive performance. Hence, monitoring and prevention of ketosis during the first 6 wk of lactation is necessary to optimize reproductive performance of lactating dairy cows.

Key words: early lactation, elevated milk BHB, Holstein cow, reproductive performance

INTRODUCTION

Dairy cows experience negative energy balance during early lactation because feed intake alone cannot support the energy demand for body maintenance as well as milk production (Wathes et al., 2007). In addition, nutrient partitioning in terms of glucose utilization is favored toward the mammary gland for milk synthesis, exacerbating the negative energy balance. The high metabolic demand during lactation-induced negative energy balance is met with by mobilization of fat from adipose tissue to release nonesterified fatty acids (**NEFA**) for energy. The NEFA can either be used as fuel by other tissues or further metabolized in the liver. The metabolic consequences of NEFA in the liver include utilization as metabolic fuel for hepatocytes, conversion to ketone bodies as alternate to glucose for other tissues and repackaging into very-low density lipoproteins as fat for systemic redistribution. However, if the NEFA load exceeds the liver's metabolic capacity, the excess fatty acids are stored leading to fatty liver. In contrast, rapid and excessive ketogenesis can lead to ketosis (Duffield et al., 2009), defined as high concentration of ketone bodies in the blood.

The prevalence of ketosis is high during the first 2 wk (Duffield et al., 1998; Walsh et al., 2007; McArt et al., 2012; Santschi et al., 2016), although the risk of ketosis lasts at least until 42 d of lactation (Mahrt et al., 2015). Depending on the presence or absence of clinical signs, ketosis is classified as clinical (**CK**) and subclinical ketosis (**SCK**), respectively. Some studies have also categorized ketosis into 2 "types" depending on the etiological background and the lactational stage when

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hyperketonemia occurs in relation to calving (Holtenius and Holtenius, 1996; Herdt, 2000; Zhang and Ametaj, 2020). Type I ketosis usually develops between 15 to 42 DIM as a result of reduced feed intake leading to lower gluconeogenesis, hypoglycemia and hypoinsulinemia, which stimulate rapid ketogenesis and lower fat accumulation in the liver (Holtenius and Holtenius, 1996). Type II ketosis usually develops in the first 14 DIM as a result of insulin resistance resulting in hyperinsulinemia, hyperglycemia and lower gluconeogenesis, which promote excessive body fat mobilization resulting in increased ketogenesis and fat accumulation in the liver (Grummer, 1993; Holtenius and Holtenius, 1996; Herdt, 2000).

The gold standard for diagnosing ketosis is measuring the concentration of the major ketone body, BHB, in blood (plasma or serum) samples. Various studies have used automated analyzers or hand-held cow-side meters to measure concentration of BHB and have used different thresholds to diagnose ketosis. The most commonly used thresholds of blood BHB concentrations are 1.2–2.9 mmol/L and 3.0 mmol/L to categorize subclinical and clinical ketosis, respectively (McArt et al., 2011; Borchardt and Staufenbiel, 2012; Suthar et al., 2013). Recent technological advances in milk analysis have shown that milk can be a noninvasive sample suited for monitoring BHB levels (Denis-Robichaud et al., 2014). Indeed, milk is routinely put through Fourier-transformed infrared (**FTIR**) testing for quality analyses including BHB measurement. Although several studies have used direct measurement techniques used for blood BHB assay, recent studies have used FTIR data to study milk BHB and ketosis in lactating cows (Benedet et al., 2019). Similar to blood BHB levels, different thresholds of milk BHB have been used to determine ketosis. However, recent studies have used, based on correlation between BHB measurement by direct and FTIR methods, the thresholds of 0.15–0.19 and $\geq$0.20 mmol/L of milk BHB as suspect and positive for hyperketonemia, respectively (Koeck et al., 2014; Santschi et al., 2016).

Several authors have indicated an association between ketosis and reproductive performance (Walsh et al., 2007; Wathes et al., 2007; Ospina et al., 2010; McArt et al., 2012; Rutherford et al., 2016). In contrast, other studies have shown that SCK and healthy cows had similar reproductive performance parameters such as first service to pregnancy (Kessel et al., 2008; Chapinal et al., 2012; McArt et al., 2012), pregnancy at first insemination (Chapinal et al., 2012), and calving to pregnancy (Wathes et al., 2003; Ospina et al., 2010; Chapinal et al., 2012; McArt et al., 2012). Although ketosis occurs in 2 different etiological background as mentioned above, there are no studies investigating the effect of late and early ketosis on reproductive performance. Thus, the objective of our study was to use FTIR milk BHB and actual breeding data from the Lactanet database to investigate associations between timing and amplitude of elevated milk BHB (**EMB**) and reproductive performance of lactating dairy cows. Because the data on circulating concentration of BHB were not available, we chose to use the following terminologies: Negative EMB, Suspect EMB and Positive EMB to classify data based on the levels of milk BHB. Because there is no clear evidence if Type I and II ketosis are associated with abnormal insulin concentration and resistance found in human diabetes, we will refer to EMB occurring during wk 1–2 of lactation as early EMB and that occurring during wk 3–6 of lactation as late EMB. Establishing the relationship between EMB "types" and reproductive performance will enhance our understanding of the effect of ketosis subtypes, which is imperative for the development of individualized strategies to optimize reproductive performance of lactating dairy cows.

MATERIALS AND METHODS

Data Management

Facility Animal Care Committee approval was not necessary for this study, as no animals were used for the sample collection and all data used for the study were part of the routine milk recording program.

The original data set obtained from Lactanet, collected between January 01, 2014, to November 30, 2016, contained records of 726,259 cows coming from 4,596 herds (median size of 137 cows) with the following information: cow identification number, herd code, parity, calving date, test-day milk BHB, 305-d milk yield, lactation end reasons (i.e., dry off, death, or sale), and breeding service records, and parental EBV for heifers (305-d milk yield, 305 d fat yield, and 305 d protein yield). To classify EMB into different types, we divided the 5–42 DIM interval into 2 periods, period 1 (5–14 DIM) and period 2 (15–42 DIM), representing, as mentioned above, the time of occurrence of early and late ketosis (Herdt, 2000). The details criteria and number of cows retained in the database at each inclusion-exclusion step are given in Figure 1A. First, we retained cows having 2 milk BHB records, one in each lactation period. Of the 33,821 cows retained, 27,095 cows had and 6,726 cows did not have a subsequent calving recorded. To analyze cows that had a reasonable breeding management, we restricted the interval of calving to first service to 20–200 DIM. With this restriction, there were 26,946 cows with a subsequent calving and 6,624 cows without a subse-

quent calving retained in the database. For cows with a recorded subsequent calving, we retained 26,517 cows with a reasonable calving interval of 300 to 600 d (Pryce et al., 2000). For these cows with a subsequent calving, the conception date was defined as the date of the last service. Gestation length was derived from the last service date and subsequent calving date and cows with a reasonable gestation length of 265 to 295 d (Olds et al., 1979) were retained, leaving 24,110 cows with reasonable calving intervals and gestation lengths in the database. From among the combined 30,734 (6,726 + 24,110) cows, we retained cows that had "lactation end reason" as normally dried or sold or died and excluded cows with no known lactation end reason. Following these inclusion and exclusion exercises, the final data set had a total of 30,413 lactating Holstein cows from 3,172 herds, with number of cows included for analysis ranging from 1 to 274 and a median of 6 cows per herd. The frequency distribution of cows during both period 1 (5–14 DIM) and period 2 (15–42 DIM) in the final data set is presented in Figure 1B. Finally, the season of calving was classified into 4 categories (fall: September to November; winter: December to February; summer: June to August, and spring: March to May).

Categorization of Milk BHB Data

The threshold levels for milk BHB concentration were establish based on the criteria outlined by the previous studies (Denis-Robichaud et al., 2014; Santschi et al., 2016): <0.15 mmol/L = negative (**NEG**); 0.15 to 0.19 mmol/L = suspect (**SUSP**); and ≥0.20 mmol/L = positive for EMB. Based on these threshold levels of milk BHB during the 2 lactation periods, cows were categorized into 7 different EMB types (Table 1). To account for the effect of previous milk yield, multiparous cows were classified into production-level categories according to their previous lactation yield and primiparous cows were classified based on their parental EBV (ECM, kg/305 d). The production level categories were low (bottom 25%), medium (middle 50%), or high (top 25%) producers. Energy-corrected milk yield of 305 d was calculated as ECM (kg/305 d) = 12.55 × fat (kg/305 d) + 7.39 × protein (kg/305 d) + 0.2595 × milk yield (kg/305 d) (Tyrrell and Reid, 1965; Santschi et al., 2016).

Reproductive Parameters Analyzed

The association of EMB with reproductive performance of lactating cows was studied by analyzing 4 major fertility performance indicators: (1) calving to first service interval, defined as number of days between calving and first service; (2) first service to conception interval, defined as number of days between first service after calving and conception; (3) days open, defined as the number of days between calving and service that resulted in a conception; and (4) calving interval, defined as number of days between 2 consecutive calvings.

Statistical Analysis

Statistical analysis was performed using SAS 9.4 (SAS Institute Inc.). Descriptive statistics were generated with the FREQ procedure of SAS. A generalized logit function was used to determine the risk factors associated with the dependent variable being EMB category (nominal categorical outcomes): NEG, EARLY_SUSP, EARLY_SUSP_Pro, EARLY_POS, EARLY_POS_Pro, LATE_SUSP or LATE_POS in early-lactation cows. The risk factors in the model included parity, previous 305-d milk yield, calving season, calving year, and region.

Kaplan and Meier (1958) estimates of the survival function were used to measure the reproductive parameters using the LIFETEST procedure of SAS. Cows that had a lactation end reason of died or sold and cows ended a lactation normally, but missed a subsequent calving record were right censored. The hazard ratios of reproductive parameters were determined using the PHREG procedure of SAS with Cox proportional hazard ratio (Cox, 1972) fitted for time calving to first service, first service to conception, days open and calving interval. The model used was

$$\lambda i(t|Xi) = \lambda 0(t)\, \exp(x_i'\beta),$$

where $\lambda i(t|Xi)$ is the hazard ratio of having the first service after calving at time t, conception after the first service at time t, conception after calving at time t (days open) or calving interval at time t; $\lambda 0(t)$ is the baseline hazard function; β is an unknown vector of regression coefficients for the covariates; x' is a vector for the fixed effect of EMB type (Table 1), parity (1, 2, 3, and ≥4), previous 305-d milk yield (low, medium, and high), calving season (fall, summer, winter, and spring), calving year (2014 to 2016), and region (1 to 18), EMB × parity, and EMB × previous 305-d milk yield, and herd as a random effect; and i is the combination of fixed effect levels for a specific observation. If statistically nonsignificant ($P \geq 0.05$), the interaction factors were manually removed by manual backward stepwise elimination. Cows that had a lactation end reason of died or sold and cows that ended a lactation normally, but missed a subsequent calving record were considered as right censored. The Cox proportional hazard model assumptions were found to be supported

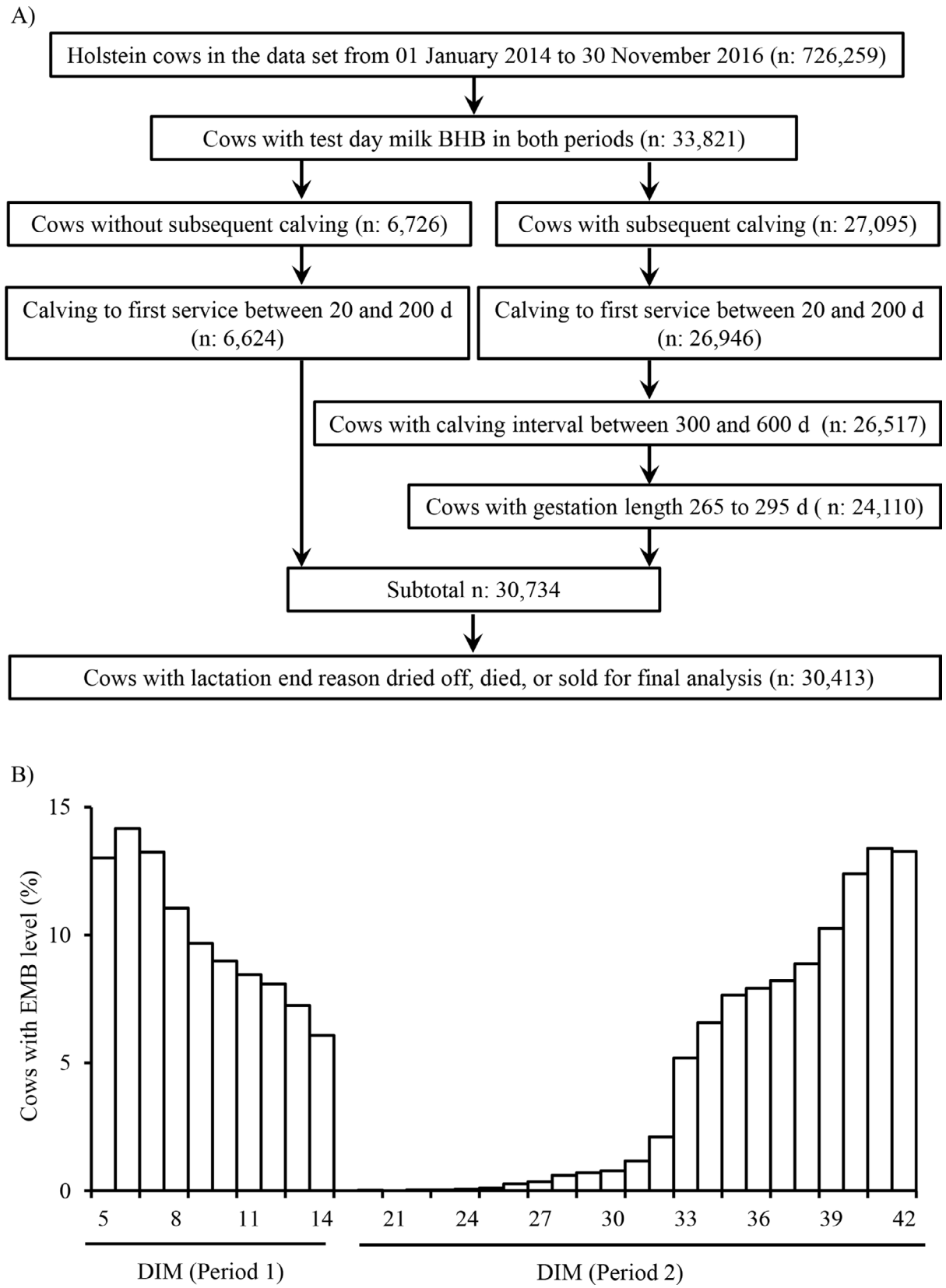


Figure 1. (A) Flowchart showing data cleaning and preparation for final analysis. (B) Distribution of cows with elevated milk BHB (EMB) level in early lactation: period 1 (5–14 DIM) or period 2 (15–42 DIM).

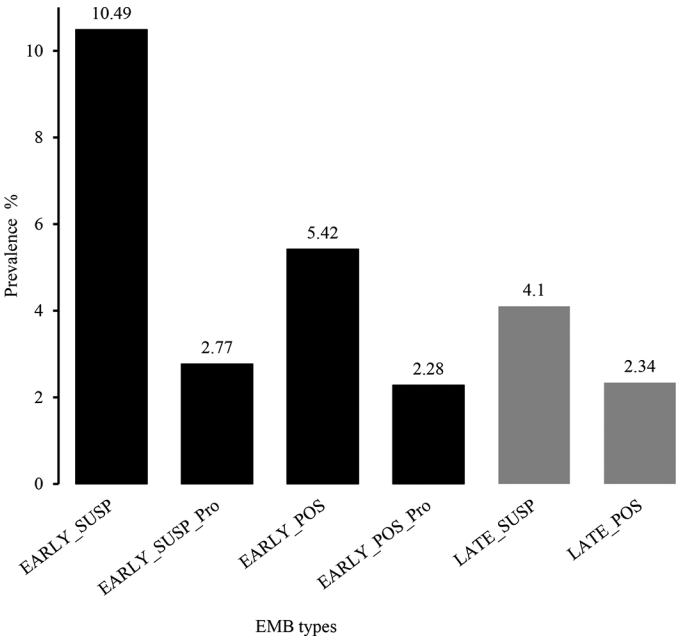


Figure 2. Prevalence (%) of elevated milk BHB (EMB) types in lactating cows. Different EMB types were based on test-day milk BHB concentrations during lactation period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM). Milk BHB concentrations were estimated from Fourier-transform infrared analyses of milk samples collected on test days. EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

by a nonsignificant relationship between Schoenfeld residuals against the survival time ($P > 0.05$).

RESULTS

After preliminary editing, the final data sets used for the analyses included a total of 30,413 cows (Figure 1A), of which 9,716 (32%) were parity 1, 8970 (29.5%) were parity 2, 6251 (20.5%) were parity 3, and 5,476 (18%) were parity ≥4. The distribution of cows with elevated milk BHB level during period 1 (5–14 DIM) or period 2 (15–42 DIM) is represented in Figure 1B. The prevalence of EMB within the first 6 wk of lactation is shown in Figure 2. The overall prevalence of EMB using a threshold ≥0.15 mmol/L of milk BHB was 27.4% within 42 DIM, with early EMB (20.96%) 3 times more prevalent than late EMB (6.44%). Of the 6 EMB types the highest prevalence was of EARLY_SUSP (10.49%). The prevalence of POS was lower than SUSP in both late and early EMB.

The differences in least squares means of test-day milk BHB concentrations during the period of 5–80 DIM among different EMB type cows were tested

Table 1. Classification of elevated milk BHB (EMB) types based on test-day milk BHB concentrations during period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM)[1]

BHB mmol/L (5 to 14 DIM)	BHB mmol/L (15 to 42 DIM)	EMB[2]
<0.15	<0.15	NEG
0.15 to 0.19	<0.15	EARLY_SUSP
0.15 to 0.19	≥0.15	EARLY_SUSP_Pro
≥0.20	<0.15	EARLY_POS
≥0.20	≥0.15	EARLY_POS_Pro
<0.15	0.15 to 0.19	LATE_SUSP
<0.15	≥0.20	LATE_ POS

[1]Milk BHB concentrations were estimated from Fourier-transform infrared analyses of milk samples collected on test days.

[2]NEG = negative; EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

using Scheffe's multiple comparison tests (Figure 3). As expected due to our grouping method, there were differences in milk BHB levels during the period of 5–42 DIM in different EMB group cows compared with NEG cows ($P < 0.05$). We included the period of 43–80 DIM to examine if the milk BHB concentrations of cows with EMB were different from those of healthy (NEG) cows during the period from the end of the ketosis-risk period (Mahrt et al., 2015) to the median day of first service after calving (Stangaferro et al., 2018). The mean milk BHB levels were well below the POS and SUSP ranges through the period from 5 to 80 DIM in NEG cows. The mean milk BHB levels during lactational period 1 (5–14 DIM) were within SUSP range in both EARLY_SUSP and EARLY_SUSP_pro groups of cows. The mean milk BHB levels during lactational period 2 (15–42 DIM) were below SUSP range in EARLY_SUSP cows, but continued to be within or above SUSP range in EARLY_SUSP_pro cows. The mean milk BHB levels during lactational period 1 were above SUSP range in both EARLY_POS and EARLY_POS_pro cows. The mean milk BHB levels during lactational period 2 were below SUSP range in EARLY_POS cows, but continued to be within or above SUSP range for EARLY_POS_pro cows. The mean milk BHB levels during lactational period 1 were below SUSP range for both LATE_SUSP and LATE_POS groups of cows, whereas the milk BHB levels during lactational period 2 were within and above SUSP range in LATE_SUSP and LATE_POS cows, respectively. Most importantly, the levels of milk BHB in all EMB type cows were similar to NEG cows ($P > 0.05$) and below SUSP range during the period from the end of the ketosis-risk period to the median day of first service after calving (43–80 DIM).

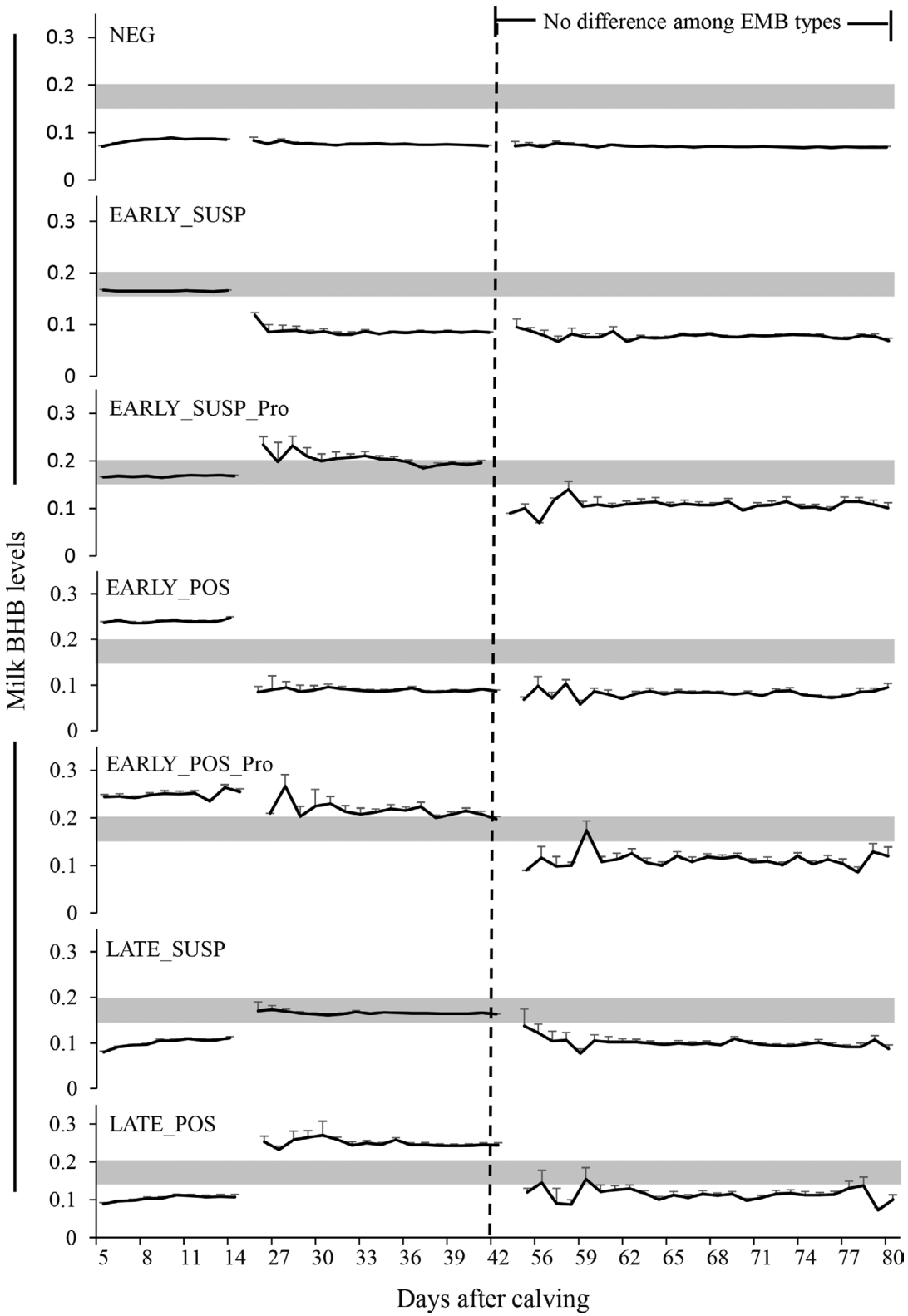


Figure 3. Daily mean milk BHB (mmol/L) of different elevated milk BHB (EMB) types in early-lactation cows. Different EMB types were based on test-day milk BHB concentrations during the period 1–80 DIM, including lactation period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM). Milk BHB concentrations were estimated from Fourier-transform infrared analyses of milk samples collected on test days. EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive. Data are presented as mean ± SEM.

Table 2. Odds ratios (OR) and 95% CI for association of parity with development of elevated milk BHB (EMB) during early-lactation cows within 42 DIM[1]

EMB type[2]	Parity[3]	OR	95% CI		*P*-value
			Lower	Upper	
EARLY_SUSP	2 vs. 1	0.79	0.71	0.87	<0.0001
EARLY_SUSP	2 vs. 3	0.76	0.68	0.85	<0.0001
EARLY SUSP	2 vs. 4	0.65	0.59	0.73	<0.0001
EARLY_SUSP	3 vs. 1	1.04	0.93	1.15	0.51
EARLY_SUSP	3 vs. 4	0.86	0.77	0.97	0.01
EARLY_SUSP	4 vs. 1	1.20	1.08	1.34	0.0007
EARLY_SUSP_Pro	2 vs. 1	0.87	0.71	1.07	0.18
EARLY_SUSP_Pro	2 vs. 3	0.44	0.36	0.54	<0.0001
EARLY_SUSP_Pro	2 vs. 4	0.39	0.32	0.48	<0.0001
EARLY_SUSP_Pro	3 vs. 1	2.00	1.62	2.37	<0.0001
EARLY_SUSP_Pro	3 vs. 4	0.89	0.74	1.07	0.22
EARLY_SUSP_Pro	4 vs. 1	2.20	1.81	2.68	<0.0001
EARLY_POS	2 vs. 1	0.70	0.61	0.80	<0.0001
EARLY_POS	2 vs. 3	0.60	0.52	0.70	<0.0001
EARLY_POS	2 vs. 4	0.50	0.43	0.59	<0.0001
EARLY_POS	3 vs. 1	1.15	1.00	1.32	0.04
EARLY_POS	3 vs. 4	0.84	0.72	0.97	0.02
EARLY_POS	4 vs. 1	1.38	1.20	1.58	<0.0001
EARLY_POS_Pro	2 vs. 1	0.91	0.72	1.15	0.42
EARLY_POS_Pro	2 vs. 3	0.40	0.32	0.50	<0.0001
EARLY_POS_Pro	2 vs. 4	0.39	0.30	0.49	<0.0001
EARLY_POS_Pro	3 vs. 1	2.23	1.85	2.81	<0.0001
EARLY_POS_Pro	3 vs. 4	0.97	0.79	1.19	0.74
EARLY_POS_Pro	4 vs. 1	2.35	1.90	2.92	<0.0001
LATE_SUSP	2 vs. 1	1.44	1.23	1.70	<0.0001
LATE_SUSP	2 vs. 3	0.77	0.66	0.90	0.001
LATE_SUSP	2 vs. 4	0.60	0.51	0.70	<0.0001
LATE_SUSP	3 vs. 1	1.89	1.58	2.22	<0.0001
LATE_SUSP	3 vs. 4	0.78	0.66	0.92	0.003
LATE_SUSP	4 vs. 1	2.40	2.03	2.84	<0.0001
LATE_POS	2 vs. 1	1.46	1.16	1.84	0.001
LATE_POS	2 vs. 3	0.53	0.43	0.65	<0.0001
LATE_POS	2 vs. 4	0.42	0.34	0.52	<0.0001
LATE_POS	3 vs. 1	2.75	2.19	3.45	<0.0001
LATE_POS	3 vs. 4	0.79	0.65	0.97	0.02
LATE_POS	4 vs. 1	3.47	2.77	4.35	<0.0001

[1]Different EMB types were based on test-day milk BHB concentrations during period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM).

[2]EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

[3]4 = ≥4.

Risk Factors for Developing EMB During Early Postpartum

We used a generalized logit model to relate explanatory variables with EMB during the risk period of ketosis. The explanatory factors of EMB included parity, energy-corrected previous 305-d milk yield, calving season, year of calving, and agricultural regions. Parity was a significant source of variation for risk with increasing parity being associated with an increased risk of developing EMB during the first 6 wk of lactation. However, parity one cows were 21% and 30% more likely to develop EMB types EARLY_SUSP and EARLY_POS, respectively, than parity 2 cows (Table 2). Milk yield in previous lactation was associated with EMB (Table 3). Cows with a medium level of 305-d milk yield in their previous lactation had 11% higher risk of EARLY_SUSP EMB than cows with high level of 305-d milk yield in their previous lactation. Cows with medium and high 305-d milk yield had 13% and 15% lower risk of developing EARLY_POS EMB, and 18% and 20% lower risk of LATE SUSP EMB, respectively than cows with a low level of 305-d milk yield in their previous lactation. However, previous lactation milk yield had no significant risks associated with EARLY_SUSP_Pro, EARLY_POS_Pro or LATE_POS EMB (Table 3). Season of calving was associated with risk of developing EMB, with higher risk of EMB types observed when calving occurred in winter (Table 4). Other risk factors which were significantly associated

Table 3. Odds ratios (OR) and 95% CI for association of previous 305-d milk yield with development of elevated milk BHB (EMB) in early-lactation cows within 42 DIM[1]

EMB type[2]	Previous 305-d milk yield[3]	OR	95% CI		*P*-value
			Lower	Upper	
EARLY_SUSP	2 vs. 1	1.01	0.92	1.11	0.83
EARLY_SUSP	2 vs. 3	1.11	1.01	1.22	0.03
EARLY_SUSP	3 vs. 1	0.91	0.81	1.02	0.11
EARLY_SUSP_Pro	2 vs. 1	1.08	0.90	1.30	0.38
EARLY_SUSP_Pro	2 vs. 3	1.08	0.91	1.29	0.35
EARLY_SUSP_Pro	3 vs. 1	1.00	0.81	1.23	0.99
EARLY_POS	2 vs. 1	0.87	0.76	0.98	0.02
EARLY_POS	2 vs. 3	1.02	0.90	1.15	0.79
EARLY_POS	3 vs. 1	0.85	0.74	0.99	0.03
EARLY_POS_Pro	2 vs. 1	0.96	0.79	1.16	0.66
EARLY_POS_Pro	2 vs. 3	1.00	0.83	1.20	0.98
EARLY_POS_Pro	3 vs. 1	0.96	0.77	1.21	0.67
LATE_SUSP	2 vs. 1	0.82	0.72	0.95	0.008
LATE_SUSP	2 vs. 3	1.03	0.89	1.19	0.68
LATE_SUSP	3 vs. 1	0.80	0.68	0.95	0.01
LATE_POS	2 vs. 1	1.06	0.88	1.29	0.53
LATE_POS	2 vs. 3	1.13	0.94	1.36	0.19
LATE_POS	3 vs. 1	0.94	0.75	1.19	0.61

[1]Different EMB types were based on test-day milk BHB concentrations during period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM).

[2]EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

[3]1 = low previous 305-d milk yield; 2 = medium previous 305-d milk yield; 3 = high previous 305-d milk yield.

with developing EMB in the model were year of calving ($P < 0.001$) and agricultural region ($P = 0.01$).

Days from Calving to First Service

The median days from calving to first service interval and the hazard ratio of first service after calving for different EMB categories and other covariate variables, obtained from the Kaplan-Meier survival and Cox proportional hazard ratio analyses, are presented in Figure 4 and Supplemental Figure S1 (https://doi.org/10.6084/m9.figshare.22718047.v1; Alemu et al., 2023). Compared with NEG cows with median interval of 73 d of calving to first service, the rate of first service was 7% lower chance ($P = 0.011$) and 2 d delay for cows classified as EARLY_POS, and with 12% of lower chance ($P = 0.005$) and 5 d delay for EARLY_POS_Pro cows for receiving their first service after calving. However, the rate of first service and the median calving to first service interval for cows in other EMB categories were not significantly different from that of NEG cows ($P > 0.05$).

Considering the effect of the other determinant factors (Figure 4), the high previous 305-d milk yield cows had a lower rate of first service ($P = 0.004$), with a median calving to first service interval of 74 d, compared with low yield cows (median 73 d). Whereas there was no significant difference in the median calving to first service interval between low and medium yield cows ($P > 0.05$). Multiparous cows had lower rate of first service after calving ($P < 0.0001$) and longer calving to first service interval (73–36 d) as compared with the first parity cows (median 72 d). The median calving to first service interval was shorter with a higher rate of first service for cows calving in the summer (median of 73 d; $P = 0.0002$) and winter (median of 72 d; $P < 0.0001$) compared with cows calving in the fall (median of 74 d). Other determinant variables which were significantly associated with interval from calving to first service after calving in the model were the year of calving ($P = 0.009$), agricultural region ($P < 0.0001$), and the random effect of herd ($P < 0.0001$).

Days from First Service to Conception

The median days of first service to conception interval and hazard ratio of conception in cows of different EMB categories and covariate variables are presented in Figure 5 and Supplemental Figure S2 (https://doi.org/10.6084/m9.figshare.22718047.v1; Alemu et al., 2023). Cows NEG for EMB had a median of 42 d interval from first service to conception, an equivalent of 2 estrous cycles. Compared with NEG cows, the rate of conception was 10% lower for EARLY_SUSP_Pro ($P = 0.013$) with a median of 55 d interval from first service to conception. EARLY_POS_pro cows had an 8% lower rate ($P = 0.049$) with a median of 48 d inter-

Table 4. Odds ratios (OR) and 95% CI for association of calving season with development of elevated milk BHB (EMB) in early-lactation cows within 42 DIM[1]

EMB type[2]	Calving season[3]	OR	95% CI		*P*-value
			Lower	Upper	
EARLY_SUSP	Spring vs. summer	0.95	0.85	1.07	0.41
EARLY_SUSP	Spring vs. winter	0.80	0.71	0.90	<0.001
EARLY_SUSP	Spring vs. fall	0.99	0.89	1.11	0.90
EARLY_SUSP	Summer vs. winter	0.84	0.75	0.94	0.002
EARLY_SUSP	Summer vs. fall	1.04	0.95	1.14	0.41
EARLY_SUSP	Winter vs. fall	1.24	1.11	1.38	<0.001
EARLY_SUSP_Pro	Spring vs. summer	1.12	0.90	1.39	0.31
EARLY_SUSP_Pro	Spring vs. winter	0.61	0.49	0.76	<0.0001
EARLY_SUSP_Pro	Spring vs. fall	0.87	0.71	1.06	0.16
EARLY_SUSP_Pro	Summer vs. winter	0.54	0.44	0.67	<0.0001
EARLY_SUSP_Pro	Summer vs. fall	0.75	0.63	0.90	0.007
EARLY_SUSP_Pro	Winter vs. fall	1.36	1.14	1.63	<0.001
EARLY_POS	Spring vs. summer	1.05	0.91	1.22	0.47
EARLY_POS	Spring vs. winter	0.89	0.76	1.04	0.15
EARLY_POS	Spring vs. fall	1.13	0.98	1.30	0.09
EARLY_POS	Summer vs. winter	0.85	0.73	0.98	0.03
EARLY_POS	Summer vs. fall	1.07	0.94	1.22	0.30
EARLY_POS	Winter vs. fall	1.27	1.09	1.45	0.001
EARLY_POS_Pro	Spring vs. summer	1.06	0.85	1.33	0.60
EARLY_POS_Pro	Spring vs. winter	0.73	0.58	0.92	0.007
EARLY_POS_Pro	Spring vs. fall	1.07	0.86	1.33	0.55
EARLY_POS_Pro	Summer vs. winter	0.68	0.55	0.85	<0.001
EARLY_POS_Pro	Summer vs. fall	1.00	0.82	1.23	0.96
EARLY_POS_Pro	Winter vs. fall	1.47	1.19	1.81	<0.001
LATE_SUSP	Spring vs. summer	1.15	0.96	1.36	0.12
LATE_SUSP	Spring vs. winter	0.77	0.65	0.93	0.006
LATE_SUSP	Spring vs. fall	0.93	0.79	1.09	0.38
LATE_SUSP	Summer vs. winter	0.67	0.57	0.80	<0.0001
LATE_SUSP	Summer vs. fall	0.81	0.70	0.94	0.006
LATE_SUSP	Winter vs. fall	1.20	1.02	1.41	0.03
LATE_POS	Spring vs. summer	1.67	1.32	2.10	<0.0001
LATE_POS	Spring vs. winter	0.89	0.71	1.12	0.33
LATE_POS	Spring vs. fall	1.04	0.85	1.27	0.71
LATE_POS	Summer vs. winter	0.53	0.42	0.68	<0.0001
LATE_POS	Summer vs. fall	0.62	0.50	0.77	<0.0001
LATE_POS	Winter vs. fall	1.16	0.94	1.43	0.15

[1]Different EMB types were based on test-day milk BHB concentrations during period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM).

[2]EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

[3]Fall = September to November; winter = December to February; spring = March to May; summer = June to August.

val from first service to conception. LATE_POS cows had an 11% lower rate of conception ($P = 0.011$) with a median of 60-d interval from first service to conception. Additionally, EARLY_POS and LATE_SUSP cows had a modestly lower rate of conception (5%, $P = 0.072$ and 6%, $P = 0.057$, respectively) with median intervals of 55 and 51-d from first service to conception, respectively. Finally, there was no difference in rate of conception and the interval from first service to conception between NEG and EARLY_SUSP cows ($P = 0.35$).

Considering the effect of the other determinant factors (Figure 5), cows with higher previous 305-d milk yield had lower rate of conception ($P < 0.01$) and longer median interval from first service to conception (44 d and 45 d for the medium and high milk yield, respectively) as compared with cows with low milk yield (median interval of 41 d). Parity had significant effect on rate of conception and first service to conception interval. Fourth and greater parity cows had 32% lower rate of conception ($P < 0.0001$) followed by third parity (24% lower; $P < 0.0001$) and second parity (18% lower; $P < 0.0001$) with longer median interval from first service to conception (73–76 d) compared with first parity cows (72 d). The median interval from first service to conception was shorter and rate of conception was higher ($P = 0.0003$) for cows calving in the summer compared with cows calving in fall, with no difference between cows calving during the fall, spring and winter seasons. Other determinant variables which

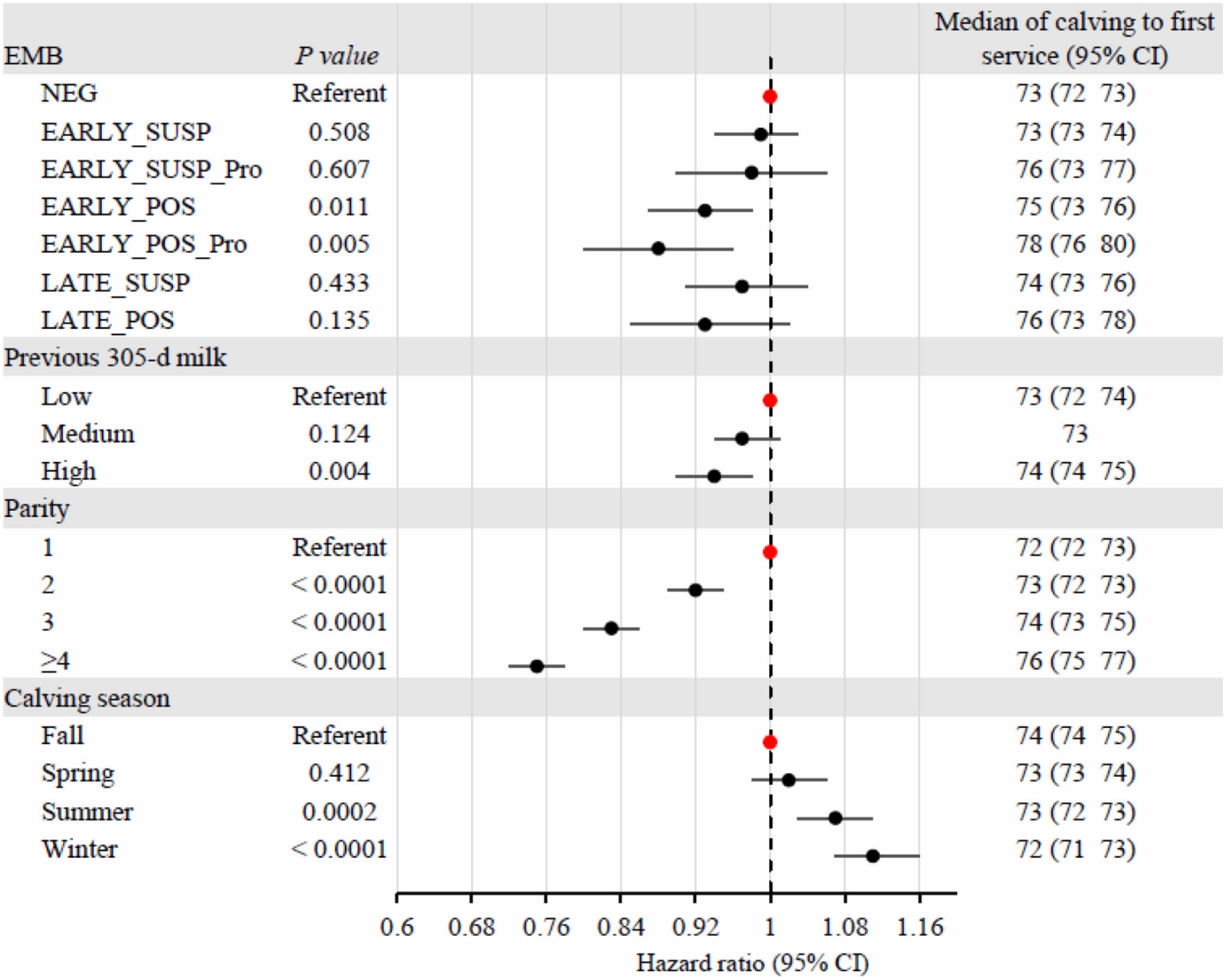


Figure 4. Estimated hazard ratio and median survival days of different elevated milk BHB (EMB) types, previous 305-d milk yield, parity, and calving season associated with days from calving to first service. Different EMB types were based on test-day milk BHB concentrations during lactation period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM). Milk BHB concentrations were estimated from Fourier-transform infrared analyses of milk samples collected on test days. EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

were significantly associated with interval from first service to conception in the model were the year of calving ($P < 0.0001$) and the random effect of herd ($P < 0.0001$), whereas agricultural region did not have a significant association ($P > 0.05$).

Days Open

The median interval of days open (calving to conception interval) and hazard ratio of conception for different EMB categories and other covariates are presented in Figure 6 and Supplemental Figure S3 (https://doi.org/10.6084/m9.figshare.22718047.v1; Alemu et al., 2023). Cows NEG for EMB had a median of 118 d days open. Rate of conception was lower for cows in all EMB categories ($P < 0.01$) except for EARLY_SUSP compared with NEG cows. The median interval for cows of all EMB categories, except for EARLY_SUSP, was 5 to 24 d longer than that of NEG cows. The rate of conception was lower with longer median days open for cows with medium ($P = 0.004$) and high ($P < 0.0001$) previous 305-d milk yield compared with low milk yield. Multiparous cows had longer days open and lower rate of conception than first parity cows ($P < 0.0001$). The rate of conception was lower for cows calving in the spring ($P = 0.002$) with longer median days, and

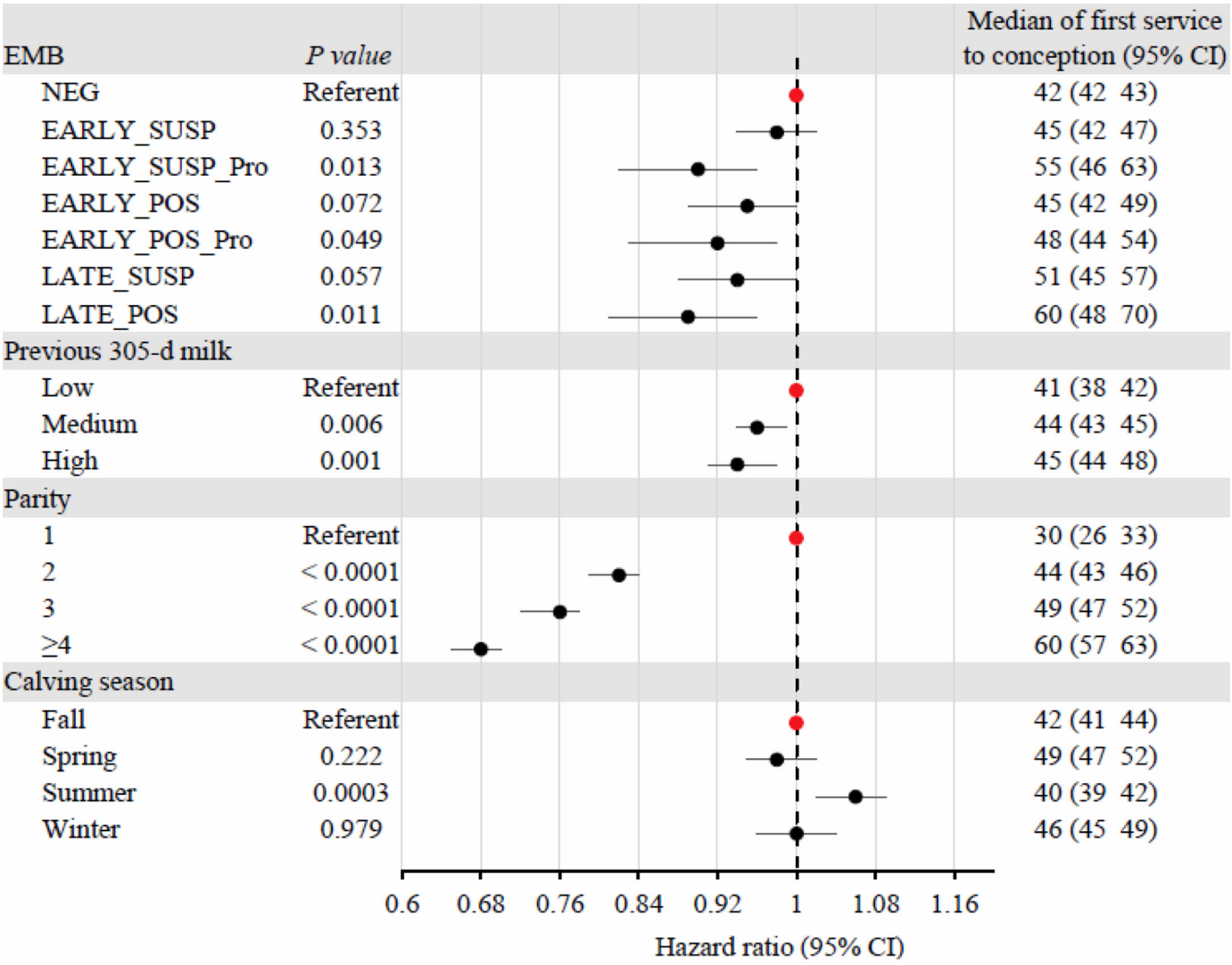


Figure 5. Estimated hazard ratio and median survival days of different elevated milk BHB (EMB) types, previous 305-d milk yield, parity, and calving season associated with days from first service to conception. Different EMB types were based on test-day milk BHB concentrations during lactation period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM). Milk BHB concentrations were estimated from Fourier-transform infrared analyses of milk samples collected on test days. EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

higher for cows calving in the summer ($P = 0.007$) with shorter median days compared with cows which calved in the fall. Other determinant variables which were significantly associated with days open in the model were the year of calving ($P < 0.0001$), agricultural region ($P = 0.016$) and the random effect of herd ($P < 0.0001$).

Calving Interval

The median days of calving interval and hazard ratio of subsequent calving for different EMB categories and other covariates are presented in Figure 7 and Supplemental Figure S4 (https://doi.org/10.6084/m9.figshare.22718047.v1; Alemu et al., 2023). Cows NEG for EMB had a median of 397 d days of calving interval. Rate of subsequent calving was lower for cows in all EMB categories ($P < 0.01$) except EARLY_SUSP compared with NEG cows. The median calving interval for cows all EMB categories, except for EARLY_SUSP, was 7–24 d longer than that of NEG cows. The rate of subsequent calving was lower with longer median calving interval in cows with medium ($P = 0.005$) and high ($P < 0.0001$) level of previous 305-d milk yield compared with low level of milk yield (Figure 7). Multiparous cows had longer calving interval and lower rate of subsequent calving interval than first parity cows ($P < 0.001$; Figure 7). Rate of subsequent calving was lower for cows calv-

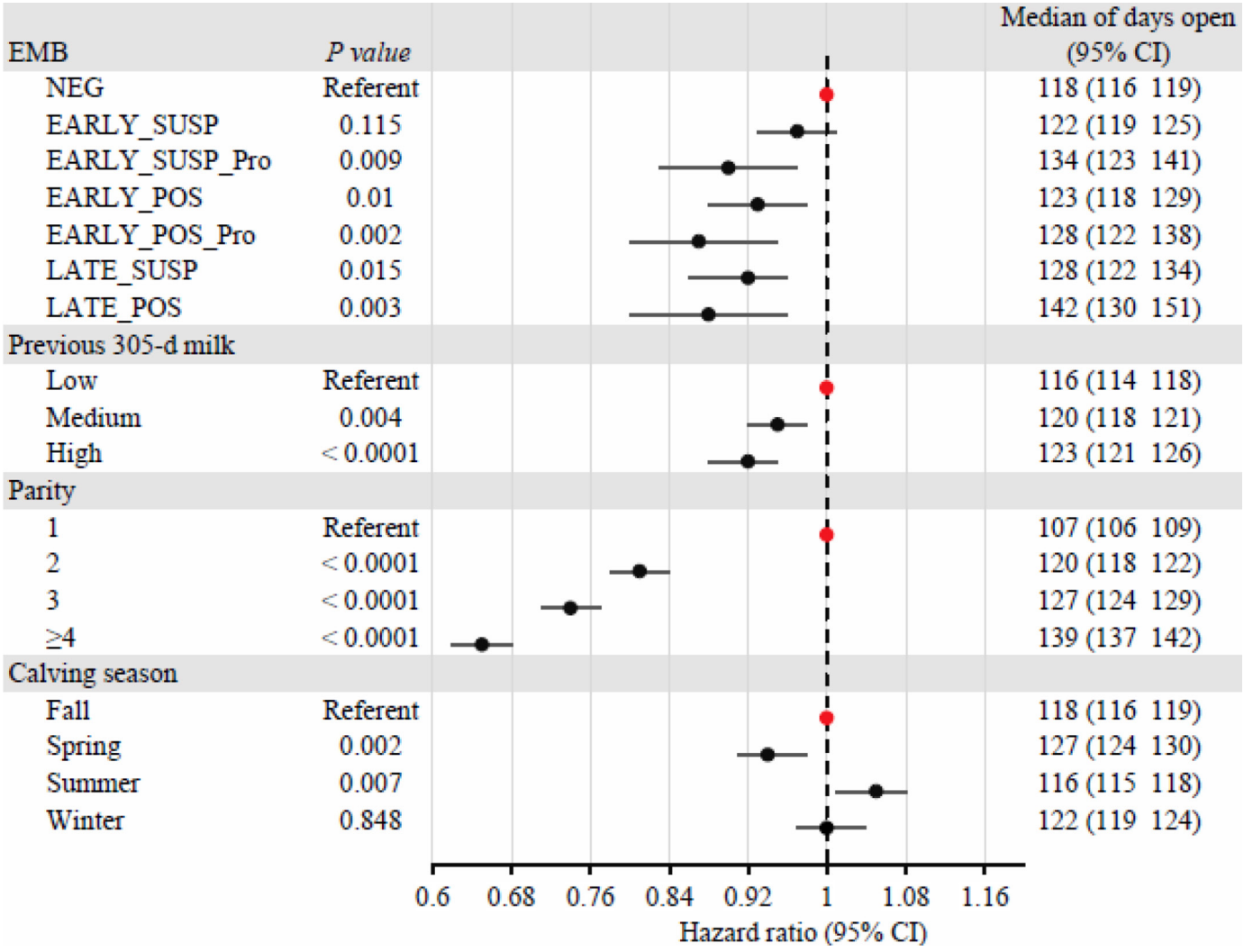


Figure 6. Estimated hazard ratio and median survival days of different elevated milk BHB (EMB) types, previous 305-d milk yield, parity, and calving season associated with days open (days from calving to conception). Different EMB types were based on test-day milk BHB concentrations during lactation period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM). Milk BHB concentrations were estimated from Fourier-transform infrared analyses of milk samples collected on test days. EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

ing in spring ($P = 0.002$) with longer median calving interval, and higher for cows calving in the summer ($P = 0.006$) with shorter calving interval compared with cows calved in fall, whereas cows calving in the winter ($P = 0.85$) had no difference (Figure 7). Other determinant variables which were significantly associated with calving interval in the model were the year of calving ($P < 0.0001$), agricultural region ($P = 0.021$), and the random effect of herd ($P < 0.0001$).

DISCUSSION

Although numerous studies have assessed the association between ketosis (clinical or subclinical) and various reproductive parameters (Walsh et al., 2007; Ospina et al., 2010; Raboisson et al., 2014; Rutherford et al., 2016), no study has examined the effect of ketosis subtypes – based on the timing and amplitude of hyperketonemia – on reproductive performance of lactating cows. Heterogeneity in the data analyzed by combining ketosis subtypes may be one of the causes of lack of association between hyperketonemia and reproductive parameters in some studies (Kessel et al., 2008; Chapinal et al., 2012; McArt et al., 2012). It is also possible that breeding and therapeutic strategies applied, if any, by the producer would have differed depending on whether ketosis is subclinical or clinical and lactation stage, thus masking the ketosis effect on reproductive performance. Therefore, we sought to examine the as-

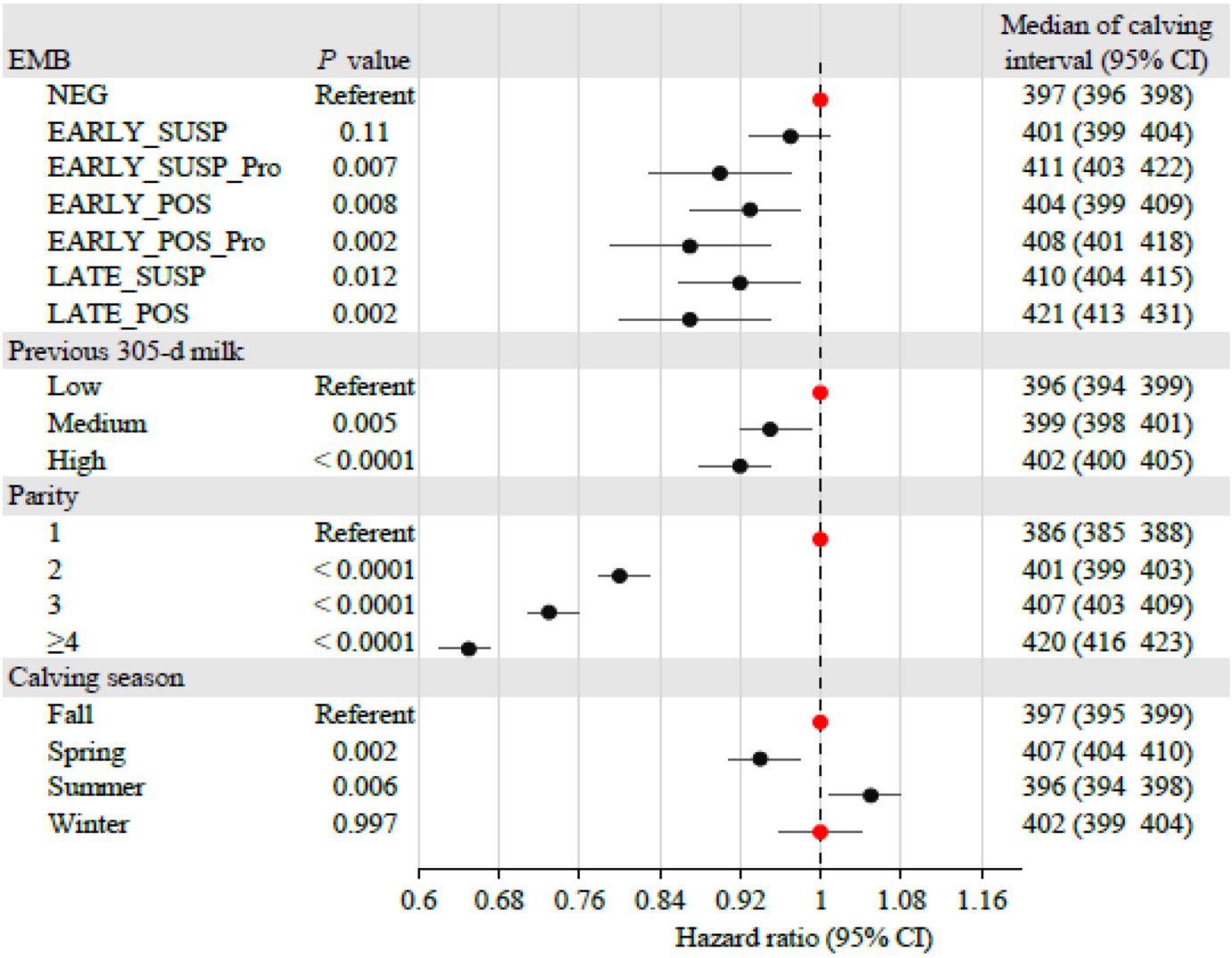


Figure 7. Estimated hazard ratio and median survival days of different elevated milk BHB (EMB) types, previous 305-d milk yield, parity, and calving season associated with calving interval. Different EMB types were based on test-day milk BHB concentrations during lactation period 1 (5 to 14 DIM) and period 2 (15 to 42 DIM). Milk BHB concentrations were estimated from Fourier-transform infrared analyses of milk samples collected on test days. EARLY_SUSP = early suspect; EARLY_SUSP_Pro = prolonged early suspect; EARLY_POS = early positive; EARLY_POS_Pro = prolonged early positive; LATE_SUSP = late suspect; LATE_POS = late positive.

sociation between individual ketosis subtypes, using the milk BHB data as an indicator, and reproductive performance indicators in this study.

Although studies have posited that early and late ketosis occur during the first 2 wk and 3–6 wk of lactation, respectively (Grummer, 1993; Holtenius and Holtenius, 1996; Herdt, 2000) there are no large-scale data summarizing the BHB level differences among the ketosis subtypes. Our data provide the summary view of milk BHB profiles among different EMB groups of cows that milk BHB levels do rise to the levels attributable to subclinical and clinical ketosis during the first 6 wk of lactation. But most importantly, this allowed us to use test-day milk BHB data to examine the milk BHB profile beyond 42 d and up to 80 d of lactation. Our data demonstrate that even though there may be a small subset of cows that present higher milk BHB levels, it is safe to assume that BHB levels in milk stabilize to healthy levels (<0.15 mmol/L) beyond 6 wk of lactation in most cows. This is supported by previous studies (Gillund et al., 2001; Mahrt et al., 2015) that showed that the occurrence of ketosis was distributed up to 42–50 d after calving.

In our study we found that the overall prevalence of EMB was 27.4% over 42 d with approximately 2-thirds of those cases categorized as early EMB occurring before 14 DIM. Among the EMB subtypes considered in this study, the highest prevalent subtype was

EARLY_SUSP (10.49%). These observations are in line with other studies reporting the highest prevalence of EMB, using milk BHB data, to be in the first 2 wk of lactation and decreasing afterward (van der Drift et al., 2012; Koeck et al., 2014; Santschi et al., 2016). Our data further provide quantitative prevalence rates beyond 2 wk up to 6 wk of lactation, beyond which the prevalence of EMB is considerably low. It is important to note that at least a quarter of the cows with EMB in the first 2 wk (early) will continue to have EMB during the next few weeks.

Although, cows with increasing parity had overall higher odds of EMB, parity one cows were more likely to develop EARLY_SUSP and EARLY_POS EMB. In line with this, first parity cows were observe to have the highest prevalence of hyperketonemia in the first week of lactation (Santschi et al., 2016). This increased odds of EMB in primiparous cows can be attributed to multiple factors including energy requirement to support their own growth, late fetal development, mammary development and the stress linked to transition from heifer to cow status. Surprisingly, we found that increased previous 305-d milk yield was associated with lower odds of some, but not with other EMB types. A previous study found that cows with higher milk yield in previous lactation had increased milk BHB concentrations at the first test day (Viña et al., 2017). The apparent difference between this study (Viña et al., 2017) and our study could be due to the fact that we did not have data to include other risk factors for higher milk BHB in our analysis, such as body condition at calving and length of dry period. Also, the study by Viña et al. (2017) considered the milk BHB on the first test-day that occurred between 5 and 35 DIM, whereas our study considered 2 test-day milk BHB values during 5–42 DIM to categorize cows into different types of EMB. In our study calving season showed an association with developing EMB types. Our observation that winter calving cows had higher risk of developing EMB compared with cows calving in fall, spring or summer, confirms the results from previous studies that the winter is a period of higher risk for ketosis (Berge and Vertenten, 2014; Vanholder et al., 2015).

The existing evidence on the effect of ketosis on reproductive parameters has been inconsistent. Some studies have shown that cows with subclinical ketosis had decreased conception rates (Wathes et al., 2003) and longer calving to first service interval and days open (Rutherford et al., 2016). Conversely, other studies have found no association between SCK and first service to pregnancy (Kessel et al., 2008; Chapinal et al., 2012; McArt et al., 2012), pregnancy rate at first insemination (Chapinal et al., 2012), and days open (Wathes et al., 2003; Ospina et al., 2010; Chapinal et al., 2012; McArt et al., 2012). The inconsistency could be explained by multiple factors such as: variation in the included covariates; different BHB concentration threshold to define ketosis; using milk or blood BHB data; and considering ketosis without subclassifications as early and late. In the present study, we used 2 test-day milk BHB data to classify cows into different EMB types and also carefully considered possible covariates, the information for which was available on the databases we used. Following systematic iterations to exclude the factors that did not have a significant effect on the reproductive parameters assessed, we included parity, previous milk yield, calving season, agriculture region of the herds and calendar year in the final model assessing the association between EMB subtypes and reproductive parameters. Inclusion of the other determinant variable allowed us to evaluate the effect of EMB types on reproductive performance over and above the effects of other variables such as parity, previous 305-d milk yield and calving season.

The calving to first service interval was longer with lower rate of first service for cows with EARLY_POS and EARLY_POS_Pro EMB, but not other EMB types. Assuming ketosis signs and comorbidities (e.g., abomasal displacements and metritis) in these 2 categories, the delayed first service could be a reflection of problems with reproductive organs and the proactive producer's decision including delaying breeding and using assisted reproductive technologies. Unfortunately, it was not possible to specify which was the reason for the delayed first service with the data available for analysis in this study. The calving to first service interval and rate of first service were not affected in cows with EARLY_SUSP and EARLY_SUSP_Pro EMB. This is in contrast with a previous study showing 15% lower rate of first service with a median 8-d delay in cows with SCK during the first or second week of lactation (Walsh et al., 2007). However, the same study (Walsh et al., 2007) found that the cows with SCK diagnoses in both of the first 2 wk had similar odds of first insemination compared with non-SCK cows. In addition, this study (Walsh et al., 2007) included all cows $\geq$1.0 mmol/L during wk 1 and cows $\geq$1.4 mmol/L during wk 2 after calving, whereas our study EARLY_SUSP and EARLY_SUSP_Pro EMB cows had 0.15–0.19 mmol/L of milk BHB (which corresponds to 1.2–2.9 mmol/L of serum BHB). These differences between our studies may explain the different observations. Also, it is intriguing that cows in LATE_SUSP and LATE_POS EMB categories did not have delayed first service.

The interval from the first insemination to conception is an important indicator of reproductive efficiency in dairy cows. It is the interval between a producer's decision to breed and conception for each cow, and a

delay in this interval can be attributed to the prolonged effects of metabolic perturbations during the early stages of lactation. Compared with NEG cows, the first service to conception interval was 3 to 12 d longer in EMB cows, except for those in EARLY_SUSP EMB group. An increase in the first service to conception interval by 3 d would mean that 1 in 7 cows would have taken one additional insemination and an additional estrous cycle of 21 d in length. Considering early EMB events occurring in the first 2 wk, our observations are in line with a previous study that the rate of pregnancy after first service was approximately 50% lower in cows classified as SCK in the first 2 wk after calving (Walsh et al., 2007). Further, our observation of longer first service to conception interval in both prolonged EMB (EARLY_SUSP_Pro and EARLY_POS_Pro) cows was 6–12 d longer than NEG cows, it is apparent that prolonged elevation of EMB is more deleterious to reproductive performance of lactating cows. Another study found that the first service to conception interval was not different between ketosis and negative groups (Kessel et al., 2008), although this study had a lower threshold of 1 m*M* BHB during a much wider period of observation of 14 wk after calving. The most important finding of our study is that the LATE_POS cows had the longest first service to conception interval. Although emphasis has been on ketosis during the first 2 wk of lactation (early) because of its high prevalence, our data demonstrate that late ketosis is equally detrimental to reproductive processes in lactating cows.

Days open is another important indicator of reproductive performance as this parameter is a combination of the calving to first service and first service to conception intervals. Compared with NEG cows, the median days open was 4 to 24 d longer in all EMB cows except for those in EARLY_SUSP EMB category. Our finding that EARLY_SUSP cows were not affected is in contrast to previous studies. Cows with elevated serum BHB in the first 2 wk of lactation had decreased risk of pregnancy after 70 d (Ospina et al., 2010), although this study included all cows with BHB $\geq$1.0 mmol/L potentially combining both CK and SCK cows. Cows with elevated serum BHB ($\geq$1.2 mmol/L) in the first week of lactation had 30% reduction in pregnancy by 150 DIM, whereas cows with elevated serum BHB in the second week did not have reduction in pregnancy compared with nonhyperketonemia cows (Rodriguez et al., 2022), although this study also potentially combined CK and SCK cows. It should be pointed out that it is not possible to rule out such mixing, although minimal, of CK and SCK cows in our study given this study is based on test-day milk samples. Another study found that among cows diagnosed as SCK during the first 2 wk of lactation, the rate of pregnancy was lower until 140 d into lactation, but not thereafter in SCK compared with non-SCK cows (Walsh et al., 2007). It should be noted that we included cows with up to 600-d calving interval indicating that cows with the potential conception day up to 320 d of lactation were included in our analysis. Finally, calving interval was also significantly longer for all EMB categories except for EARLY_SUSP. As calving interval is the sum of days open and gestation length, it was also prolonged by 4 to 24 d in affected EMB categories compared with NEG cows, with LATE_POS cows having the longest median calving interval.

In line with the literature (Gröhn and Rajala-Schultz, 2000; Grimard et al., 2006), cows of second parity or greater had significantly lower hazard ratio of calving to first service, conception to first service, days open and calving interval than cows of first parity. Our finding that all reproductive parameters analyzed were negatively affected by the level of the previous 305-d milk production concurs with a previous study showing high producing cows required more services and longer time to pregnancy than low producing cows (Berger et al., 1981). The observation that cows that calved during summer had shorter first service to conception, days open and calving intervals compared with cows calving in fall was unexpected and in contrast to a previous report that summer calving cows had lower rate of pregnancy than fall calving cows (Farin et al., 1994). The reproductive parameters analyzed in this data set were also influenced by the agriculture regions, which is attributable to changing feed quality and climate year-to-year in different regions.

The readers are cautioned that the factors that could affect reproductive performance, such as body condition score, peripartum morbidities, hormone levels, could not be included in the statistical model due to unavailability of such information. Given the expected norm of 25–30 d gap between test-day milk collection, there were fewer cows with test-day collection during wk 3 and 4. Because they represented field conditions of variable frequency of test-day collections by the producers and exclusion of such cows did not influence our statistical inferences, those cows were retained in the analyses presented in this study. More specific categorization of EMB, and meticulous inclusion and exclusion of data from more than 30-thousand cows have enabled us to provide compelling evidence that EMB occurring at different stages of early lactation can be deleterious to reproductive performance of dairy cows. Late ketosis occurring between wk 3 and 6 of lactation appears to have the strongest adverse effects on the fertility of lactating dairy cows as LATE_POS cows had the longest first service to conception interval, days open and calving interval. Separating EARY_SUS EMB from other

categories revealed that it may not have a drastic negative effect on cow fertility as reproductive parameters of first service to conception interval, days open and calving interval were not significantly different between NEG and EARLY_SUSP cows. Overall, the results of this study show that dairy producers should pay attention to EMB during the transition period to avoid its negative effect on reproductive performance of their cows.

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